

ILYA PAPOU

8 years in IT · 5+ in DevOps / SRE

I develop agent infrastructure and orchestration for software engineering, with MCP/A2A integration and human-in-the-loop workflow design. I build Kubernetes platforms and shared DevSecOps release pipelines used in production by multiple customers. I enjoy working with AWS and Google Cloud and prefer them for cloud projects.

PROFESSIONAL EXPERIENCE

SIRENA-TRAVEL — SENIOR SRE / PLATFORM ENGINEER

Mar 2025 — Sep 2026 · Contract · Thailand · Remote

- Built and operated a bare-metal RKE2 platform with encrypted etcd and mTLS for inter-DC Kafka, service-to-service traffic, PostgreSQL and Redis. Managed certificates centrally in Vault and propagated updates using External Secrets Operator (ESO), webhooks and Ansible.
- Standardized service delivery through GitLab CI/CD, shared Helm charts and Argo CD.
- Owned on-call response, investigated failures and documented recovery and prevention steps.
- Centralized images and dependencies in Nexus. Enforced SAST, GitGuardian and Trivy gates blocking releases and Argo CD sync to production. Assigned developers tasks for vulnerability and dependency fixes, secret removal and development password rotation.
- Integrated kagent and Mattermost for AI-assisted engineering, with agent definitions and platform configuration managed through IaC and GitOps.

SBERBANK INSURANCE BROKER — HEAD OF DEVOPS

Jun 2023 — Mar 2025 · Full-time · Russia · Remote

- Built and led a DevOps team with 4 direct reports. Coordinated work across a wider infrastructure group of ~10 people. Established shared standards, documentation and knowledge sharing.
- Provisioned ~8 Kubernetes clusters (20+ nodes total) with Terraform and GitOps. Integrated Cloud.ru Advanced (Huawei Cloud technology) services with VMware on bare-metal servers at Cloud.ru.
- Built a Jenkins DevSecOps pipeline for 30–40 microservices: Jira-triggered releases with shared Helm charts, security checks, tests and approvals. Build and delivery took ~5 min for >6 services. Full releases with cross-department approvals completed within a day. The pipeline remained in use and supported new services after my departure.
- Built a Keycloak SSO PoC with Nginx. Coordinated microservice changes and handed off requirements and technical specs to Security and Development. Introduced Vault, encrypted service traffic, centralized monitoring and tested disaster recovery.

I-TECO — SENIOR DEVOPS ENGINEER

Apr 2022 — Jun 2023 · Full-time · Russia · Remote

- Managed development and test infrastructure on ~10 Hyper-V / Windows Server hosts, each running ~50 VMs, including Linux guests and shared engineering services.
- Built Kubernetes on Alt Linux 9.2 in an air-gapped environment. Prepared releases on USB drives for security-controlled physical transfer and deployed them using scripts.
- Maintained Airflow, Cassandra, RabbitMQ, PostgreSQL / Patroni, Docker Swarm, TeamCity and Nexus. Automated operations with Ansible, Python and Bash and used OpenVPN for isolated access.

FREELANCE / FLANT — DEVELOPMENT & DEVOPS

Nov 2019 — Mar 2022 · Selected projects · Remote

- Delivered cloud infrastructure and web applications across freelance and Flant projects using AWS (S3, Lambda, Kubernetes and IAM) and Yandex Cloud.
- Built React / Next.js and Node.js / Express applications. Deployed Java / Spring workloads to Kubernetes with Helm, GitHub Actions and Ansible.
- Supported delivery and operations with Prometheus, Grafana, ELK and PostgreSQL / Patroni.

MTS GROUP — INFORMATION SECURITY INTERN

Feb 2018 — Nov 2019 · Internship · Belarus · Remote

- Configured Linux, including Kali Linux, and analyzed boot flow and core OS components. Evaluated major virtualization models and built isolated security-testing labs.
- Mapped networks and assessed exposed services with Nmap and reconnaissance tools. Identified vulnerabilities and validated findings in controlled Metasploit exercises.
- Configured and tested VPN, Tor and I2P traffic routing. Encrypted and decrypted binary files.
- Used cryptocurrency wallets and transactions to study blockchain validation and data flow.



CONTACT

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SPOKEN LANGUAGES

Russian: Native

English: B1

Spanish: A1

German: A1

CORE STRENGTHS

Platform ownership

Production reliability

Delivery automation

Team leadership

ENVIRONMENTS

Cloud & bare metal

Isolated / air-gapped

Product & consulting

Remote teams

DOMAIN EXPERIENCE

Finance & insurance

Aviation & travel

iGaming / anti-fraud

Public sector



YOUROWN.CHAT – AGENT INFRASTRUCTURE

Personal R&D · Research / PoCs: 2025–2026 · Core development: Aug 2026–Present

R&D platform designed to coordinate developers and AI agents in a persistent conversation, preserving context across agent handoffs and long-running workflows, with human-in-the-loop review and approvals for code and infrastructure changes.

- Deployed YourOwn.Chat components on Google Cloud. Developed four reusable Helm workload profiles for services, workers, jobs and stateful applications, with schema constraints and contract tests for networking, volumes and persistence.
- Built a Go Agent Host used with YourOwn.Chat on Google Cloud: enrollment, native/Docker execution, outbound TLS/gRPC, Substrate worker registration, short-lived access and slot lifecycle control. Implemented Go adapters for official Codex and Claude binaries with A2A streaming, session continuation and cancellation. Provider login stays with the official clients.
- Defined declarative agents with Harness, ModelConfig and AgentTemplate, model/effort selection and separate dev/prod namespaces. Prepared Restricted Pod Security and default-deny ingress/egress policies with a DNS baseline.
- Designed orchestration with kagent for agent discovery and lifecycle, Agent Host for coding execution, and Temporal for durable workflows, retries and human approval waits. Designed A2A-to-workflow handoff with structured task/evidence transfer, idempotent requests, queued messages and fallback to direct execution if workflow startup fails.
- Implemented a Go policy API with strict validation, canonical request hashes, fail-closed decisions and bounded audits, documented with OpenAPI and an ADR. Added MCP relay tool allowlists and digest-pinned OCI skills for managed runtimes.
- Built and ran image and Helm-chart release workflows with MCP and agents. Prepared a Pub/Sub-triggered Cloud Build path through private staging and immutable Artifact Registry repositories, with image scanning, digest pinning and GCS release evidence linking source commits, artifacts and SBOM/provenance. Prepared Cloud Deploy promotion of the same artifacts from dev to prod after approval.
- Validated Agent Host contracts with simulated Codex/Claude providers. Tested release scanning and promotion policies.

Selected code: github.com/pilprod/yourown-chat · github.com/pilprod/kagent · github.com/pilprod/mattermost

SELECTED TECHNOLOGIES

Platform & delivery

Linux · Kubernetes · RKE2 · Docker
Terraform · HCP Terraform Stacks
Helm · Argo CD · Ansible
GitLab CI/CD · Jenkins
GitHub Actions · Nexus

Cloud infrastructure

AWS · GCP
S3 · Lambda · EKS · IAM
GKE · Cloud SQL · GCS
Artifact Registry · Cloud Build / Deploy
IAM · WIF · Secret Manager
Cloud KMS · Pub/Sub · Cloudflare

Software engineering

Go · Python · Bash · JavaScript
React · Next.js · Node.js · Express
Java · Spring · C++
OpenAPI · gRPC
Schema validation · contract tests

Security & reliability

Vault · ESO · Keycloak · mTLS · Kyverno
GitGuardian · Trivy
Prometheus · VictoriaMetrics
Grafana · ELK · OpenTelemetry
Tailscale · OPNsense

ML & data infrastructure

Airflow · Slurm · JupyterHub · MLflow
PostgreSQL · Patroni · Redis
Kafka · RabbitMQ · Cassandra
MinIO · VectorDB · BigQuery

Agent infrastructure & R&D

kagent · agentgateway
MCP · A2A · RAG
Temporal · Human-in-the-loop
LangChain · ADK
Codex · Claude Code · Gemini
vLLM · Langfuse

HOME AEROPONICS – IOT AUTOMATION

2024 – 2025 · Personal project

- Built a home aeroponic system with ESP32 / Arduino controllers and Raspberry Pi. Wrote C++ firmware for light sensors and JSON data exchange over MQTT.
- Implemented thermostat and humidistat logic for climate and humidity control, automatic watering, and relay control for fans, pumps, valves and other mechanical components.
- Used Mosquitto, Zigbee2MQTT, Home Assistant and Docker for automation, monitoring and alerts.
- Automated Raspberry Pi setup with Ansible and prototyped pH / TDS / EC monitoring.

PERSONAL R&D – ZERO-TRUST MESH · 2025

- Built an encrypted Tailscale lab mesh connecting devices and private networks across cloud, bare-metal and homelab sites in multiple geographic regions.
- Managed all mesh ACLs through GitOps, with distinct user and service-account permissions for DNS discovery and service access.
- Tested OPNsense as an NGFW for subnet routing, network segmentation and traffic filtering.